

PCCST501 COMPUTER NETWORKS — COMPREHENSIVE QUESTION BANK

Aligned with KTU Exam Pattern (Part A: 3-Mark & Part B: 9-Mark Questions) | Prof. Anju Markose,
VJCET

MODULE 1: Overview of Internet & Application Layer

Part A (3 Marks Each)

- 1. Differentiate between packet switching and circuit switching with respect to resource allocation.
- 2. Define transmission delay and propagation delay. State the formulas for both.
- 3. Why is HTTP called a stateless protocol? How are cookies used to maintain session state?
- 4. Differentiate between persistent and non-persistent HTTP connections.
- 5. Why does FTP use two separate parallel TCP connections? Name their default port numbers.
- 6. Distinguish between recursive and iterative DNS query resolution with suitable sketches.
- 7. Explain the role of Type A, Type CNAME, and Type MX DNS resource records.
- 8. How does BitTorrent achieve fast content distribution using the Rarest First piece selection policy?

Part B (9 Marks Each)

- 1. (a) Explain the ISO-OSI 7-layer reference model in detail, highlighting the functions and PDUs of each layer. (6 Marks)
(b) Illustrate the encapsulation and decapsulation process as data traverses through the protocol stack. (3 Marks)
- 2. (a) Describe the architecture and protocol flow of the File Transfer Protocol (FTP). Differentiate between Active and Passive FTP modes. (5 Marks)
(b) Explain the components and message exchange in Electronic Mail (SMTP, POP3, IMAP4, and MIME). (4 Marks)
- 3. (a) Explain the Domain Name System (DNS) architecture, server hierarchy, and DNS caching mechanism in detail. (5 Marks)
(b) Describe the Peer-to-Peer (P2P) file distribution paradigm. Explain the Tit-for-Tat and Optimistic Unchoking algorithms in BitTorrent. (4 Marks)

MODULE 2: Transport Layer & Network Layer

Part A (3 Marks Each)

- 1. Differentiate between UDP and TCP protocols based on reliability, connection state, and header overhead.
- 2. Explain the 3-Way Handshake mechanism for TCP connection establishment with sequence numbers.
- 3. What is Silly Window Syndrome? How is it resolved by Nagle's and Clark's algorithms?
- 4. Distinguish between TCP Tahoe and TCP Reno on receiving 3 duplicate ACKs.
- 5. Differentiate between select() and poll() I/O multiplexing system calls in Linux.
- 6. What is the Count-to-Infinity problem in Distance Vector routing? How does Split Horizon resolve it?
- 7. Differentiate between Leaky Bucket and Token Bucket traffic shaping algorithms.
- 8. Compare the base header features of IPv4 and IPv6.

Part B (9 Marks Each)

- 1. (a) Explain the TCP Congestion Control mechanism in detail, explaining Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery phases with a cwnd vs RTT graph. (6 Marks)
(b) Explain the calculation of the 16-bit UDP checksum using pseudo-header. (3 Marks)
- 2. (a) An organization is granted the block 192.168.10.0/24. Subnet this block to create 4 subnets. Find the subnet mask, network address, broadcast address, and usable host range for each subnet. (5 Marks)
(b) Explain the working of Dijkstra's Link-State Routing Algorithm with an example network graph. (4 Marks)
- 3. (a) Explain the implementation of the Linux Kernel routing table (fib_table) and route cache. Provide the syntax to add and delete routes using the `ip route` command. (5 Marks)
(b) Describe Multicast Routing concepts, IGMP, and compare DVMRP with PIM-SM routing protocols. (4 Marks)

MODULE 3: Data Link Layer, Ethernet & Wireless LANs

Part A (3 Marks Each)

- 1. Explain the Bit Stuffing and Byte Stuffing framing mechanisms in Data Link Control.
- 2. Compare the maximum throughputs of Pure ALOHA and Slotted ALOHA protocols.
- 3. Why is collision detection (CSMA/CD) not feasible in wireless networks? What protocol is used instead?
- 4. Derive the minimum frame size equation in Ethernet CSMA/CD networks.
- 5. Explain the working and packet formats of Address Resolution Protocol (ARP).
- 6. How does a Learning Switch build its forwarding table dynamically?
- 7. Explain the Hidden Terminal and Exposed Terminal problems in IEEE 802.11 Wireless LANs.
- 8. What is Triangle Routing in Mobile IP? How is it optimized?

Part B (9 Marks Each)

- 1. (a) A bit stream 1101011011 is transmitted using the CRC generator polynomial $G(x) = x^4 + x + 1$. Calculate the transmitted frame FCS. Show how the receiver detects errors. (5 Marks)
(b) Explain the 7-bit Hamming Code generation and error correction for data word 1011. (4 Marks)
- 2. (a) Explain the IEEE 802.3 Standard Ethernet frame format in detail. Discuss the evolution from 10 Mbps to Gigabit and 10-Gigabit Ethernet. (5 Marks)
(b) Explain Virtual LANs (VLAN - IEEE 802.1Q) and explain how VLAN tagging isolates broadcast domains. (4 Marks)
- 3. (a) Explain the IEEE 802.11 MAC sublayer operations including CSMA/CA, NAV, Inter-Frame Spaces (SIFS, PIFS, DIFS), and RTS/CTS handshake. (5 Marks)
(b) Explain the implementation of packet capturing using Linux raw sockets with SOCK_PACKET and PF_PACKET. (4 Marks)

MODULE 4: Network Management & Physical Layer

Part A (3 Marks Each)

- 1. List the key components of the SNMP Network Management architecture.
- 2. Differentiate between SNMPv1, SNMPv2c, and SNMPv3 with respect to security and performance.
- 3. Explain the Type-Length-Value (TLV) encoding in ASN.1 Basic Encoding Rules (BER).
- 4. State Nyquist's Bit Rate formula and Shannon's Channel Capacity theorem.
- 5. Differentiate between Manchester and Differential Manchester line coding techniques.
- 6. Explain Quadrature Amplitude Modulation (QAM). How many bits/ baud does 16-QAM carry?

- 7. Compare Synchronous TDM and Statistical TDM multiplexing techniques.
- 8. Compare Single-Mode and Multi-Mode optical fibers.

Part B (9 Marks Each)

- 1. (a) Explain the SNMP protocol operations (GetRequest, GetBulkRequest, SetRequest, Trap) and the MIB-2 tree hierarchy with Object Identifiers (OIDs). (5 Marks)
(b) Explain the ASN.1 data structure definition and encode an example INTEGER and OCTET STRING using BER TLV format. (4 Marks)
- 2. (a) Calculate the theoretical maximum capacity of a noisy telephone channel having a bandwidth of 4 kHz and a Signal-to-Noise Ratio (SNR) of 30 dB using Shannon's theorem. If we use 16 signal levels, what is the Nyquist bit rate? (5 Marks)
(b) Draw the digital signal waveforms for the bit sequence 01001110 using: (i) NRZ-L, (ii) NRZ-I, (iii) Manchester, (iv) Differential Manchester, (v) Bipolar AML. (4 Marks)
- 3. (a) Explain Frequency Division Multiplexing (FDM), Wavelength Division Multiplexing (WDM), and Time Division Multiplexing (TDM) in detail. (5 Marks)
(b) Discuss the physical characteristics, advantages, and limitations of Twisted Pair, Coaxial Cable, and Optical Fiber guided media. (4 Marks)